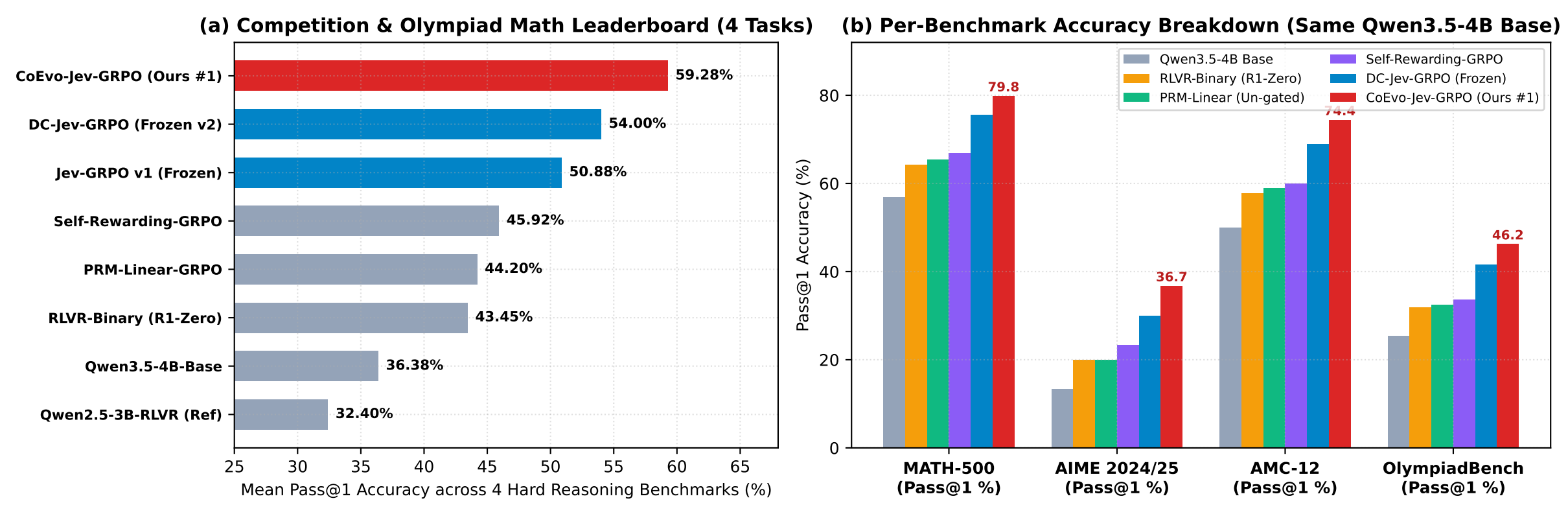
(a) Competition & Olympiad Math Leaderboard (4 Tasks)
CoEvo-Jev-GRPO (Ours #1)
DC-Jev-GRPO (Frozen v2)
Jev-GRPO v1 (Frozen)
Self-Rewarding-GRPO
PRM-Linear-GRPO
RLVR-Binary (R1-Zero)
Qwen3.5-4B-Base
Qwen2.5-3B-RLVR (Ref)
59.28%
54.00%
50.88%
45.92%
44.20%
43.45%
36.38%
32.40%
25
30
35
40
45
50
55
60
65
Mean Pass@1 Accuracy across 4 Hard Reasoning Benchmarks (%)
(b) Per-Benchmark Accuracy Breakdown (Same Qwen3.5-4B Base)
Qwen3.5-4B Base
RLVR-Binary (R1-Zero)
PRM-Linear (Un-gated)
Self-Rewarding-GRPO
DC-Jev-GRPO (Frozen)
CoEvo-Jev-GRPO (Ours #1)
Pass@1 Accuracy (%)
0
20
40
60
80
79.8
36.7
74.4
46.2
MATH-500 (Pass@1 %)
AIME 2024/25 (Pass@1 %)
AMC-12 (Pass@1 %)
OlympiadBench (Pass@1 %)